

Harshit Rohatgi

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Professional Summary

Principal software engineer and technical leader with 15+ years of experience architecting and delivering highly scalable, business-critical distributed systems across enterprise software, cloud platforms, e-commerce, and AI-enabled products. Experienced in solving ambiguous technical problems, defining long-term architecture, leading cross-functional execution, and mentoring engineering teams. Recent work focuses on enterprise AI agents, semantic data access, federated query systems, natural-language-to-SQL, type systems, and machine-learning-driven engineering workflows.

Core Skills

Programming Languages: Java, TypeScript, Python, C++

Cloud and Distributed Systems: AWS, Microservices, Distributed Systems, Federated Architectures, Low-Latency Systems

AI and Machine Learning: Large Language Models, Retrieval-Augmented Generation, Natural-Language-to-SQL, LangGraph, PyTorch, TensorFlow, scikit-learn, Azure AutoML

Data and Platform Engineering: Data Fabric, Federated Query Engines, Semantic Models, Entity Relationships, Query Planning, Telemetry, Reliability Engineering

Leadership: Technical Strategy, Architecture Reviews, Cross-Functional Leadership, Mentorship, Project Execution

Professional Experience

UiPath

Sep 2025–Present

Principal Software Engineer

Bengaluru

- Architected and delivered the integration of UiPath Data Fabric with the Agents Unified Runtime, enabling enterprise AI agents to discover, read, and update governed business entities through a unified tool interface. Drove implementation across the platform SDK, LangGraph-based agent runtime, and agent orchestration framework.
- Defined the architecture for exposing federated enterprise data as a native agent capability, allowing agents to query systems of record without data replication or movement. Led cross-team integration that achieved 72% end-to-end natural-language-to-SQL execution accuracy at a 2-second P95 inference latency.
- Established semantic and type-system foundations for reliable agent interactions with enterprise data, including entity relationships, choice sets, and read/write operation contracts.
- Shaped the technical strategy for agentic data access by addressing schema grounding, entity grain, query planning, telemetry, and production-readiness challenges across Data Fabric, agent platform, and federated query teams.
- Defined performance, reliability, and quality metrics for agent-generated queries and enterprise data operations, improving observability and operational readiness for production deployments.

Microsoft

Sep 2021–Aug 2025

Lead Software Development Engineer

Bengaluru

- Architected and delivered a mission-critical Formula Remediation Engine that automated the resolution of Excel formula incompatibilities, unblocking a strategic business deliverable affecting 12,000 tenants and 2.3 million files per month.
- Led a team of eight engineers through architecture, implementation, validation, and production rollout of the remediation platform.

- Led a five-member engineering team focused on improving Excel Online reliability and reducing behavioral gaps between Excel Online and Excel Desktop within the Excel calculation stack.
- Pioneered a generative AI workflow for validating complex, high-dimensional Excel formula features. Developed a test-case generation system grounded in the Excel calculation engine, increasing remediation accuracy to 99.9%.
- Moderated the Excel India Development Center Design Review Panel and helped improve architectural review quality, consistency, and engineering standards.

Amazon

Feb 2020–Aug 2021

Senior Software Development Engineer

Bengaluru

- Served as lead developer in the Amazon Sellers organization, building seller-verification capabilities across Amazon and AWS storefronts.
- Led development of a business process management platform built on AWS Step Functions.
- Participated in the Samurai architecture group, providing design consultation on self-contained web components, data-intensive applications, and AWS service-development practices.

Amazon Web Services

Apr 2015–Feb 2020

Software Development Engineer II

Seattle

- Served as a core engineer for AWS Marketplace across multiple layers of the platform and transaction pipeline.
- Contributed to the AWS Marketplace web rendering platform, with emphasis on federated frontend architecture and developer tooling for product teams.
- Bootstrapped an experimentation platform for AWS, with AWS Marketplace as the first customer. Implemented a low-latency service for vending recognition tokens from request identifiers, achieving 10 ms P95 latency.
- Led development of container-product support for AWS Marketplace for the AWS re:Invent 2018 launch.
- Built tooling that enabled AWS services to onboard to the AWS Marketplace platform and create service-specific marketplaces using shared platform capabilities.

Amazon Webstore

Sep 2011–Apr 2015

Software Development Engineer

Hyderabad

- Developed seller-facing tooling for the Amazon Webstore platform.
- Architected a new site-design platform using independently deployable web components and defined the initial architecture for theme design and seller-tool integration.
- Developed an auto-categorization system for seller inventory with multiple distributed integrations.
- Improved reliability and maintainability by building distributed tracing tools that exposed failures and performance issues across integration points.

Deloitte India Consulting Pvt. Ltd.

Jun 2010–Sep 2011

Business Technology Analyst

Hyderabad

- Worked in Deloitte Consulting's Information Management service line and developed data-visualization proof of concepts using the Google Maps API.